
Homework 2

MAT 3510 Due Date: Oct. 7, 2026

submit your homework to

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Instruction: Read the instructions carefully. Your homework should follow the requirements as mentioned in the syllabus.

Problem 1 [20 points]

(1) Write Matlab code to perform Gaussian elimination with LU decomposition for a linear system $Ax = b$ with $A \in \mathbb{R}^{d \times d}$, $x \in \mathbb{R}^d$, and b^d . [That is, the function you write should work for general d -dimensional square linear system.]

(2) Test your code with the following example:

$$\begin{cases} x_1 - 2x_2 + 3x_3 - 4x_4 = 4 \\ x_2 - x_3 + x_4 = -3 \\ x_1 + 3x_2 + x_4 = 1 \\ -7x_2 + 3x_3 + x_4 = -3 \end{cases}$$